GOVERNMENT DEGREE COLLEGE FOR WOMEN (AUTONOMOUS)

Begumpet, Hyderabad-500016

DEPARTMENT OF STATISTICS

Project work

On

"TO ANALYZE GOLD PRICE BY ARIMA MODEL USING R PROGRAMMING"

Done by:

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CERTIFICATE

This is to certify that the project work entitled **"TO ANALYZE GOLD PRICE BY ARIMA MODEL USING R PROGRAMMING"** is a bonafide work done by students.

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Of **B.SC(MSDS) 3rd year under the guidance of Ms.M.Anusha**, Assistant Professor of Statistics at Government Degree College For Women(A), Begumpet in partial fulfillment for the award of **B.Sc** Degree by **Osmania University, Hyderabad.**

Lecture Incharge **External Examiner**

DECLARATION

Chepur Nandika, Kuna Rasagnya, Narsapuram Harshitha, Puli Shakshitha, Rudrakshala Kavya Sri the bachelor of science students of Government Degree College for Women(A), Begumpet and affiliated to Osmania University, Hyderabad, hereby declare that the project titled **"TO ANALYZE GOLD PRICE BY ARIMA MODEL USING R PROGRAMMING"** is a genuine work of us. We have done it as a part of our B.SC curriculum under the guidance of Ms.M.Anusha, Assistant Professor of Statistics.

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1. INTRODUCTION

We're trying to figure out what the price of gold might be in the future. Gold is super valuable and people have been fascinated by it for thousands of years. We're using sophisticated statistics such as Time Series , ARIMA and a Programming language R to forecast the price of gold.

What is Gold?

Gold, a symbol of wealth, power, and beauty, has captivated human civilization for millennia. Its allure transcends cultures and civilizations, making it one of the most coveted commodities in history. In the modern era, the quest to predict gold prices has gained paramount importance, driving financial decisions, investment strategies, and economic policies. This project endeavors to explore the intricate dynamics of gold price prediction through the lens of time series analysis and ARIMA modeling.

Gold's significance spans across various domains, from ancient civilizations adorning their temples and tombs with this precious metal to modern economies relying on it as a hedge against inflation and uncertainty. Its intrinsic value, scarcity, and universal acceptance make it a unique asset class with enduring appeal. Understanding the factors influencing gold prices and accurately predicting its future trajectory is essential for investors, policymakers, and market analysts alike.

History of Gold Price:

Gold has been valuable to people for a really long time. Thousands of years ago, ancient civilizations like the Egyptians and the Greeks thought gold was special. They used it to make shiny things like jewelry and coins.

Over time, as people started trading with each other, they used gold to pay for things. Gold became a symbol of wealth and power. The more gold you had, the richer you were.

But the price of gold wasn't always the same. Sometimes, when people really wanted gold, its price went up. Other times, when people didn't care as much about gold, its price went down.

There were also big events in history that affected the price of gold. Wars, discoveries of new gold mines, and changes in how money worked all made the price of gold go up or down.

For example, during the Gold Rush in the 19th century, when people discovered gold in California and Australia, the price of gold went up because there was a lot of demand for it.

In recent times, gold prices have been influenced by things like changes in the economy, inflation, and uncertainty in the world. When people are worried about the economy or there's political instability, they often buy gold because they see it as a safe investment.

Today, gold is still valuable, and its price continues to change based on what's happening in the world. People still buy gold as a way to protect their money and as a symbol of wealth.

Understanding the history of gold prices helps us see how valuable gold has been to people throughout history and why it's still important today.

Why do Gold Price Prediction Matters?

Gold price prediction matters because it empowers stakeholders to anticipate market trends, mitigate risks, and seize opportunities. Whether it's forecasting short-term fluctuations or long-term trends, accurate predictions enable investors to optimize their portfolios, governments to formulate effective policies, and businesses to manage their finances prudently.

Predicting gold prices allows investors to capitalize on market inefficiencies, identify arbitrage opportunities, and hedge against inflation and currency devaluation. For governments and central banks, gold price forecasts inform monetary policies, reserve management strategies, and international trade

negotiations. Moreover, businesses in the mining, jewelry, and manufacturing sectors rely on gold price predictions to make informed production and pricing decisions.

How Gold Prices Have Changed Over Time

Gold prices have gone up and down a lot over the years. Sometimes it's because people really want gold, and other times it's because of things happening in the world, like wars or new discoveries of gold. Looking at how gold prices have changed in the past can help us guess what might happen in the future.

Why Predicting Gold Prices is Important

Predicting gold prices helps investors, governments, and businesses make better decisions. It's like being able to see into the future and knowing what's going to happen next in the world of money. If we can predict gold prices accurately, we can make smarter choices about things like investing or buying and selling gold.

Time Series Analysis

Time series analysis is a specific way of analyzing a sequence of data points collected over an interval of time. In time series analysis, analysts record data points at consistent intervals over a set period of time rather than just recording the data points intermittently or randomly. However, this type of analysis is not merely the act of collecting data over time.

What sets time series data apart from other data is that the analysis can show how variables change over time. In other words, time is a crucial variable because it shows how the data adjusts over the course of the data points as well as the final results. It provides an additional source of information and a set order of dependencies between the data.

Time series analysis typically requires a large number of data points to ensure consistency and reliability. An extensive data set ensures you have a representative sample size and that analysis can cut through noisy data. It also ensures that any trends or patterns discovered are not outliers and can account

for seasonal variance. Additionally, time series data can be used for forecasting—predicting future data based on historical data.

Time Series and Stationarity

When performing time series analysis, one of the most typical assumption is that the data must be stationary. If a time series' statistical attribute remains constant over an extended period of time, it is said to as stationary. Put simply, it indicates that the mean and variance of the data series are constant. Time has no effect on the data's covariance. We need the data series to be stationary in order to perform modelling. Not all series including non-stationary data are stationary by nature, but by applying various transformations, non-stationary series can be converted to stationary series.

They are given in this order: -

1) Logarithm

2) Differentiation: After d times of differencing, the time series becomes stationary; this is known as integrated of order d, or I(D). Said to be I (0) if a series becomes stationary. The series is said to be I (1) if, after computing the first difference in time, it becomes stationary.

ARIMA Model

Since the development of advanced statistical software packages, **ARIMA**, as it is more often known, has become a popular time series forecasting technique for short-term forecasts. The four main steps in developing an **ARIMA model** are **identification, estimation, diagnostics** and **forecast**. These four steps involve identifying the tentative model parameters using graphs ACF and PACF, determining the coefficient and identifying the likely model, validating the model, and using basic statistics and confidence intervals to assess the forecast's validity and monitor the model's performance. The ARIMA model breaks down historical data into two categories: **Integrated (I)** Indicates trends that are polynomial or linear, and A **moving average (MA)** that is weighted over previous errors is

displayed. As a result, it has three model parameters: **AR(p), I(d)**, and **MA(q)**, which together create the complete **ARIMA(p, d, q)** model, where

p is the **autocorrelation** order

d is the **integration** order (differencing)

q is the **moving averages** order

A non-seasonal stationary time-series can be modeled as a combination of the past values and the errors which can be denoted as ARIMA (p, d, q) or can be expressed as

$$X_t = \theta_0 + \varphi_1 X_{t-1} + \varphi_2 X_{t-2} + \quad + \varphi_p \quad X_{t-p} + \quad e_t \quad -\theta_1 e_{t-1} \quad - \quad \theta_2 e_{t-2} \quad - \quad \theta_q e_{t-q}$$

(1)

Initial differences step can be applied one or more times to eliminate the non-stationarity.

To cross check the stationarity of time series, use Augmented Duckey-Fuller test. Rejecting the null hypothesis suggests that a time series is stationary.

ARIMA models are using any of the following models are as follows:

ARIMA (1, 0, 0); Autoregressive (1)

ARIMA (0, 1, 0): Integrated model (1)

ARIMA (0, 1, 0) without constant is called random walk => Yt=a+Yt-1+et

ARIMA (0, 1, 0) with constant is called random walk with drift => Yt=a+Yt-1+et

ARIMA (0, 0, 1): moving average (1)

ARIMA (0, 0, 0): white noise model

ARIMA (0, 1, 2): Damped Holt's model

ARIMA (0, 1, 1): Basic experimental smoothing model (without constant)

ARIMA (0, 2, 2): Double exponential smoothing model

Akaike Information Criteria (AIC) is used to determine the appropriate order of a non-seasonal ARIMA model. An AIC criterion provides the approximate models towards the reality of the situation.

For a good model we have to minimize the values of AIC, Bayesian Information Criteria(BIC) used to find the perfect fit.

2. OBJECTIVES OF THE STUDY

In today's world gold plays a vital role and the demand of the gold has increased a lot .Moreover wherever we go the first question will be whether the gold price will be increasing or decreasing and in future how much it would be increasing or decreasing. So to solve this kind of problems we came with the solution. So, here are our objectives include:

Objectives of the study:

- Forecasting the future gold price by implementing ARIMA model
- To analyze the trend and pattern of gold stock through Time Series
- To analyze gold price using R

Time Series

Time series analysis is a statistical technique used to analyze sequential data points collected over time. It involves identifying patterns, trends, and seasonal variations within the data to make predictions about future values. In this study the time series is used to find the trends and relationship between the variables and to draw meaningful insights from it.

ARIMA Model

ARIMA (AutoRegressive Integrated Moving Average) model is a powerful forecasting tool commonly used in time series analysis. It combines autoregressive (AR), differencing (I), and moving average (MA) components to capture temporal dependencies and stochastic patterns within the data. In our study ARIMA model is especially used to forecast the future gold price through time series analysis using trend and seasonality.

3. METHODOLOGY

Data Collection: Gathering historical gold price data from January 2014 to January 2024 through the secondary sources.

Data Preparation: Preparing the data for further analysis to ensure consistency and accuracy.

Time Series Analysis: Analyzing the data using time series methods to identify patterns and trends.

ARIMA Modeling: Implementing ARIMA (AutoRegressive Integrated Moving Average) model to forecast gold prices.

Model Evaluation: Assess the performance of the ARIMA model using appropriate evaluation metrics.

Visualization: Visualize the predicted gold prices and compare them with actual values.

4. DATA

For our case we have collected the data from most popular website Kaggle, where we have extracted the goldstock.csv dataset of India that summarizes the gold price outlook for 10 years period from January 19, 2014 to January 22, 2024. This file consists of 2532 records and 6 variables of important financial information such as date, closing price, volume and opening price, up down.

Here you can have look on our dataset:

	Date	Close	Volume	Open	High	Low
0	1/19/2024	2029.3	166078	2027.4	2041.9	2022.2
1	1/18/2024	2021.6	167013	2009.1	2025.6	2007.7
2	1/17/2024	2006.5	245194	2031.7	2036.1	2004.6
3	1/16/2024	2030.2	277995	2053.4	2062.8	2027.6
4	1/12/2024	2051.6	250946	2033.2	2067.3	2033.1
5	1/11/2024	2019.2	270624	2029.4	2056.1	2017.3
6	1/10/2024	2027.8	183625	2036	2046.2	2025.9
7	1/9/2024	2033	194610	2034.5	2048.6	2031.9
8	1/8/2024	2033.5	199325	2052.6	2053.3	2022.7
9	1/5/2024	2049.8	208861	2051.4	2071.1	2030.8
10	1/4/2024	2050	125573	2049.3	2058.1	2043.3
11	1/3/2024	2042.8	214132	2067.9	2074.3	2038.3
12	1/2/2024	2073.4	149926	2072.7	2088.1	2064.3
13	12/29/2023	2071.8	102741	2076.1	2084.1	2067.6
14	12/28/2023	2083.5	126411	2089.5	2098.2	2074.6
15	12/27/2023	2093.1	124021	2079.3	2095.8	2072.8
16	12/26/2023	2069.8	76492	2066	2080.5	2065.1
2522	2/5/2014	1256.9	122164	1254.2	1274.5	1251.8
2523	2/4/2014	1251.2	86988	1257.9	1260.7	1246.8
2524	2/3/2014	1259.9	120942	1242	1266.1	1240.4
2525	1/31/2014	1239.8	111168	1244	1254.8	1238.2
2526	1/30/2014	1242.5	155053	1267.2	1267.8	1237.5
2527	1/29/2014	1262.2	115459	1255.8	1270.7	1248.3
2528	1/28/2014	1250.5	81426	1254.9	1261.9	1248
2529	1/27/2014	1263.5	63419	1269.9	1280.1	1252
2530	1/24/2014	1264.5	34998	1264.3	1273.2	1256.9
2531	1/23/2014	1262.5	41697	1235.1	1267.1	1230.8
2532	1/22/2014	1238.6	80262	1240.5	1243.5	1235.5

5. DATA ANALYSIS AND INTERPRETATION

Here is some factors which impacts the gold price, these are the factors which we found in our dataset. To understand these variables follow the below descriptions:

Date: For the ten-year period between January 19, 2014, and January 22, 2024, this column shows the precise calendar date that corresponds to each trading day that was recorded.

Volume: This column, which shows the total quantity of gold units purchased, defines the trading activity of gold on every day that is recorded.

Open: This column, which shows the opening price of gold for each trading day and provides an insight into the initial value at the start of trade.

Close: This column, which shows the closing price of gold for each trading day and provides an insight into the closing value at the end of trade.

High: This column provides information on the peak valuation by listing the highest price that gold hit throughout each trading day.

Low: This column shows the lowest price for gold that was recorded throughout each trading day, providing a quick look at the lowest valuation.

Loading the Data

- Importing required packages and loading the data

```
[4] #fetching required packages
    library(dplyr)
    library(readr)
    #read data
    gold_stock<-read_csv("goldstock.csv")

    New names:
    • `` -> `...1`
    Rows: 2511 Columns: 7
    ── Column specification ────────────────────────────────────────────────────────
    Delimiter: ","
    dbl  (6): ...1, Close, Volume, Open, High, Low
    date (1): Date

    ℹ Use `spec()` to retrieve the full column specification for this data.
    ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

- Fetching six records to analyze the structure of the dataset

```
[7] #Have a look at the structure of the data
    head(gold_stock,6)
```

A tibble: 6 × 7

...1	Date	Close	Volume	Open	High	Low
<dbl>	<date>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
0	2024-01-19	2029.3	166078	2027.4	2041.9	2022.2
1	2024-01-18	2021.6	167013	2009.1	2025.6	2007.7
2	2024-01-17	2006.5	245194	2031.7	2036.1	2004.6
3	2024-01-16	2030.2	277995	2053.4	2062.8	2027.6
4	2024-01-12	2051.6	250946	2033.2	2067.3	2033.1
5	2024-01-11	2019.2	270624	2029.4	2056.1	2017.3

- Arranging the data into ascending order

```
[8] #We are converting the data into ascending order
    gold_stock<-gold_stock %>% arrange(Date)
    head(gold_stock)
```

A tibble: 6 × 7

...1	Date	Close	Volume	Open	High	Low
<dbl>	<date>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
2532	2014-01-22	1238.6	80262	1240.5	1243.5	1235.5
2531	2014-01-23	1262.5	41697	1235.1	1267.1	1230.8
2530	2014-01-24	1264.5	34998	1264.3	1273.2	1256.9
2529	2014-01-27	1263.5	63419	1269.9	1280.1	1252.0
2528	2014-01-28	1250.5	81426	1254.9	1261.9	1248.0
2527	2014-01-29	1262.2	115459	1255.8	1270.7	1248.3

- Since we are forecasting the daily closing price we are extracting the **Date** and **Close** variables

```
#To create a forecast model for the daily closing gold price, we'll just be extracting the "Date" and "Close" variables.
gold_stock<-gold_stock %>% select(Close)
colnames(gold_stock)<-c("Price")
#The data includes 2,511 days of "Close" pricing history.
NROW(gold_stock)
```

```
2511
```

Exploratory Data Analysis (EDA)

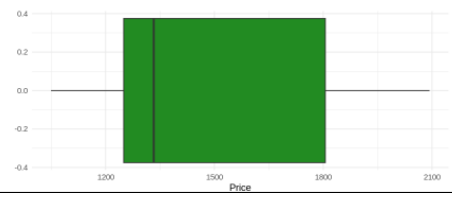
- Checking Basic Statistics

```
#Basic Statistics
summary(gold_stock)
```

```
     Price
 Min.   :1050
 1st Qu.:1250
 Median :1333
 Mean   :1499
 3rd Qu.:1806
 Max.   :2093
```

- Drawn Boxplot of **"Price"** variable to check the outliers, but no outliers were found.

```
[29] #Baxplot of Price Variable
     library(ggplot2)
     g_s1<-gold_stock %>% ggplot()+geom_boxplot(aes(x=Price),fill="forestgreen")+theme_minimal()
     options(repr.plot.width=7,repr.plot.height=3)
     #First of all, there have been no outliers found
     #Secondly, we can observe that the highest number, which was approximately 2,100, was noticeably higher than the median, which was around 1,300.
     print(g_s1)
```




```
[ ] NROW(gold_stock)
```

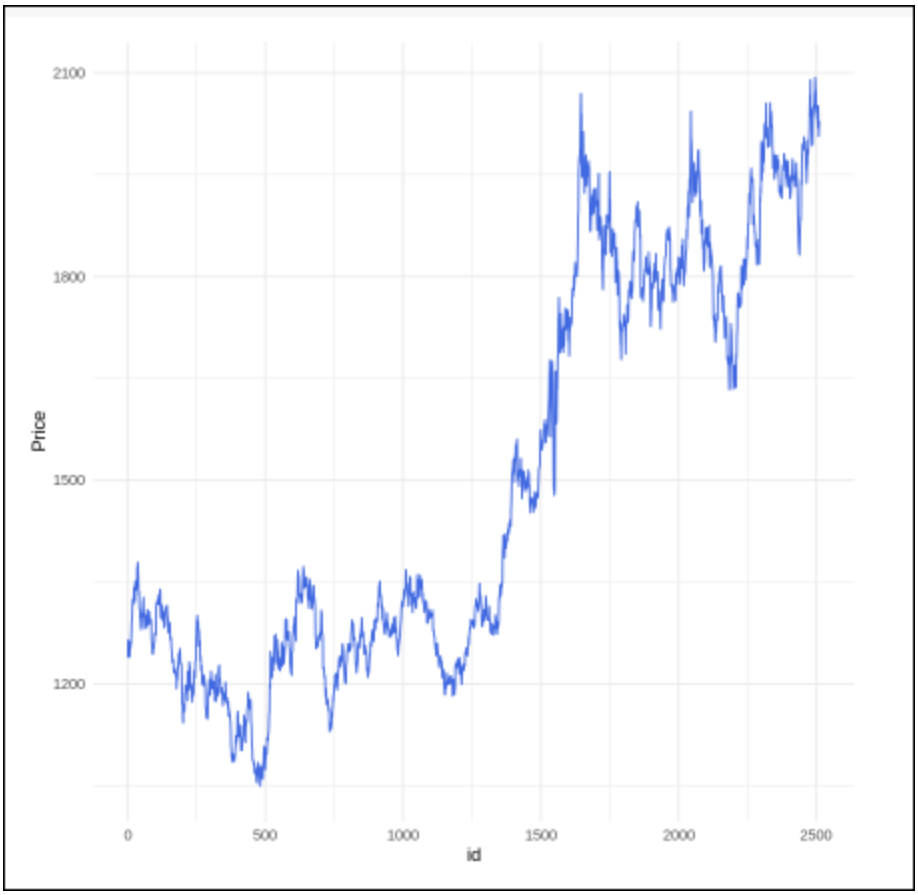
```
2511
```

```
[ ] NROW(seq(1:NROW(gold_stock)))
```

```
2511
```

- To analyze the trend of gold stock over last ten years, we have drawn a trendplot.

```
#Let's draw trendplot
id<-seq(1:NROW(gold_stock))
g_s2<-gold_stock %>% mutate(id=id) %>% ggplot()+geom_line(aes(x=id,y=Price), col="royalblue")+theme_minimal()
options(repr.plt.width=14,repr.plot.height=7)
#The price has slowly went up over the last decade, with a surge after the pandemic due to worldwide uncertainty.
#Seasonality and trend are obvious, indicating the need to make the price stationary for better forecast.
print(g_s2)
```

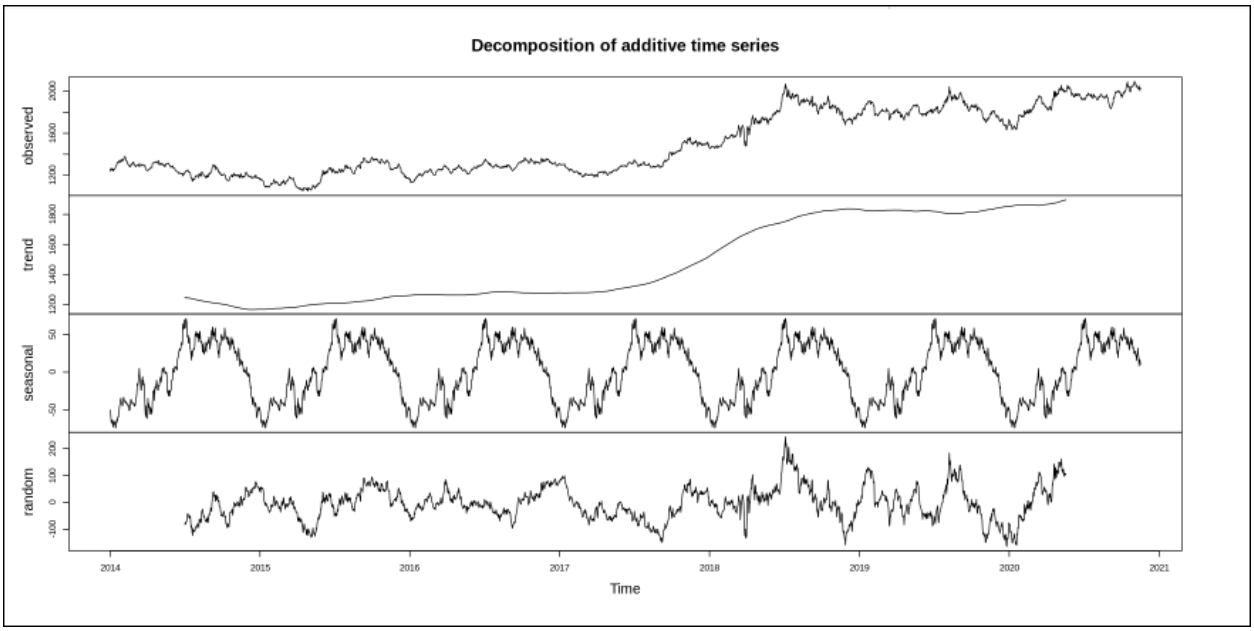


Interpretation:

From the above graph, the time series data of gold price in India from Jan 2014 to Jan 2024 shows an **upward trend**. That means the price of gold is increasing continuously especially from 1250th id , we can observe that until 1250th id the prior data is fluctuating upwards and downwards but from 1250th id onwards the price gold is increasing rapidly which indicates the data is **non-stationary**.

- **Decomposition of additive time series**

```
#To decompose time series data, convert it to 'ts' format
#For daily data, set 'freq' to '365'(a few dates have been omitted, but for convenience, we'll only use the daily data)
gold_ts<-ts(gold_stock$Price,start=c(2014,1),freq=365)
options(repr.plot.width=14,repr.plot.height=7)
#As we looked at previously, a clear "trend" and "seasonal" characteristic is observed
plot(decompose(gold_ts))
```



Interpretation:

This graph shows that the code decomposes the gold price data into four main components: **observed, trend, seasonal patterns,** and **random fluctuations**. It helps us understand the underlying patterns in gold prices, helping in accurate forecasting of future trends and fluctuations. So, we can see the trend is going upward which says the gold price increase, and seasonality revealed the pattern every year in mid time the price of gold is rising to peak and end of the year it is decreasing and we can observe some random fluctuations in random component which is due to noise or randomness in the data.

Data Preparation

- Out of 2511 records, we have trained 80% data and remaining data will be for testing

```
#Out of 2511 days,we are going to train 80% of the data
#And the remaining 20% will be for testing
print(NROW(gold_ts))
print(NROW(gold_ts)*0.8)
```

```
[1] 2511
[1] 2008.8
```

```
#The time period has been shortened as shown below because we omitted skipped dates
#But it makes no difference for the purpose of the daily price forecast merely.
print(start(gold_ts))
print(end(gold_ts))
```

```
[1] 2014    1
[1] 2020  321
```

- Splitting the data as before 183rd day of 2019 for training and remaining data for testing.

```
# we'll split the time period prior to and following the 183rd day of 2019
NROW(window(gold_ts,end=c(2019,183)))
```

```
2008
```

- Splitting the data for training and testing

```
#Splitting the data for traing and testing
train<-window(gold_ts,end=c(2019,183))
test<-window(gold_ts,start=c(2019,184))
print(NROW(train))
print(NROW(test))
```

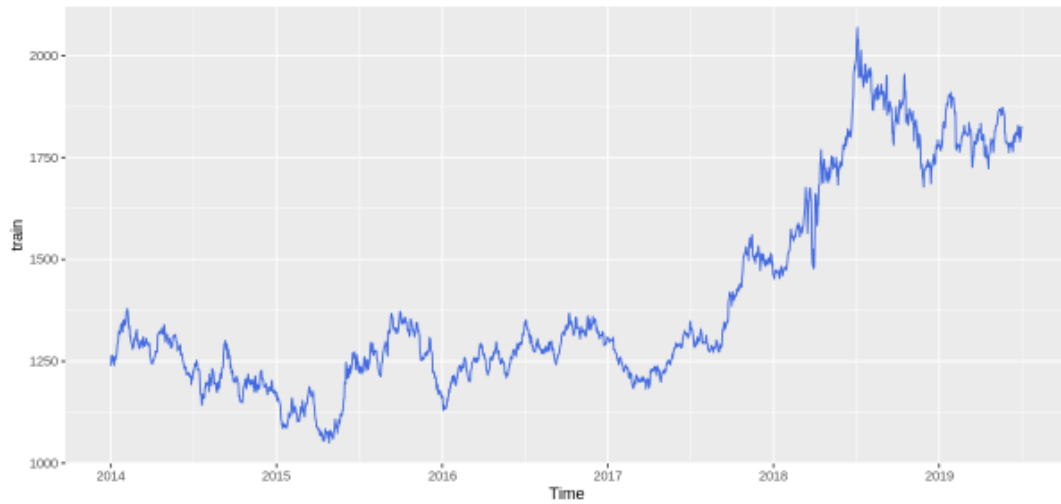
```
[1] 2008
[1] 503
```

ARIMA Model

- Installing the forecast package and have exclude before creating ARIMA model

```
install.packages('forecast', dependencies = TRUE)
#We must first exclude "trend" from the time series data before building the ARIMA model.
library(forecast)
options(repr.plot.width=10,repr.plot.height=5)
autoplot(train,col="royalblue")
```

```
Installing package into '/usr/local/lib/R/site-library'
(as 'lib' is unspecified)
```



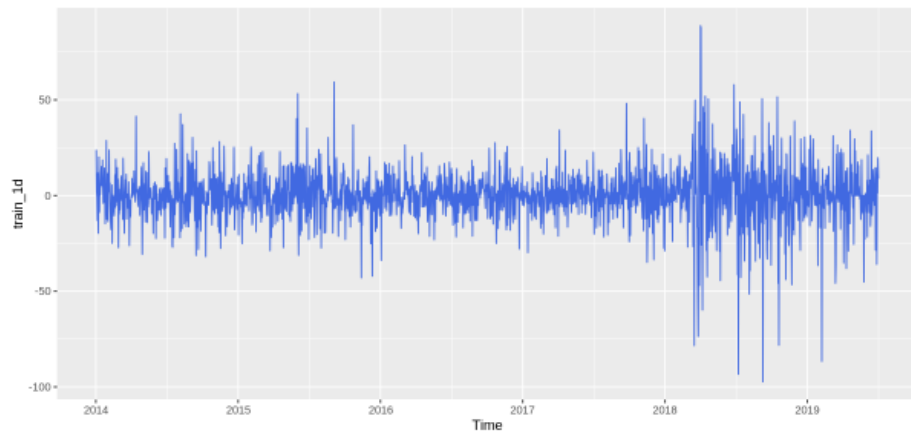
Interpretation:

The above graph we have already observed earlier which shows an **upward trend** so now in order to create an **ARIMA model** we must exclude the trend and should convert the data into stationary data.

- Applying first difference to the train data for stationarity

```
[21] #Typically, the first degree of difference will be sufficient,
     #but we'll double check using the forecast::The ndiffs() method
     print(ndiffs(train))
     #We'll use the train data's first difference.
     train_1d<-diff(train)

     [1] 1
```

```
#Plotting indicates that the "trend" has been effectively eliminated,
#but certain days have very large variances
#For these reasons, we'll return to the original data and use BoxCox Transformation.
autoplot(train_1d,col="royalblue")
```



Interpretation:

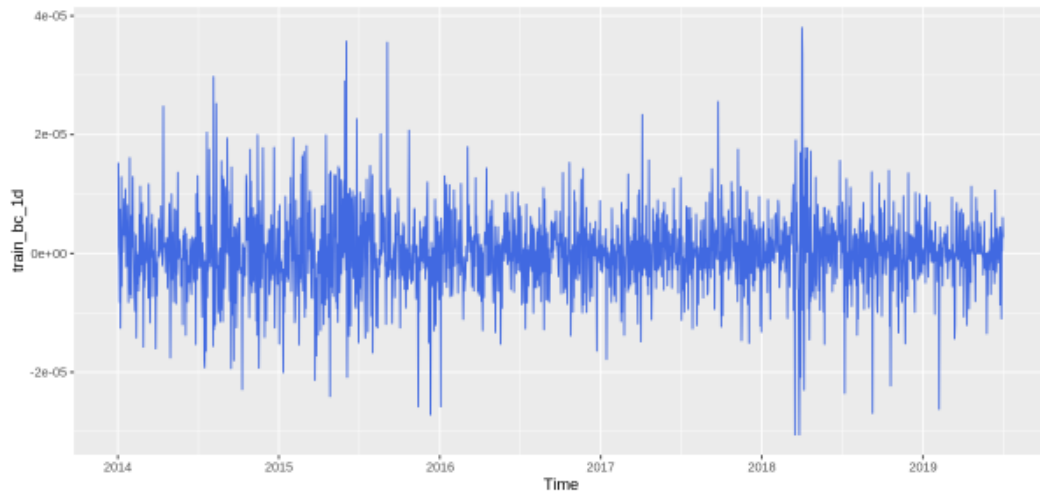
From the above graph, we can observe that after applying first difference our data became stationary. But in certain days we found huge variances. So,to make entire data into stationary we will be applying the BoxCox transformation to the original data.

- Applying BoxCox

```
[25] #We will use only the train data to get the best lambda value,
     #and then we will modify the test and train data according to that value
     train_lambda<-BoxCox.lambda(train)
     train_bc<-BoxCox(train,lambda=train_lambda)
     test_bc<-BoxCox(test,lambda=train_lambda)
```

- Applying first difference as before for the BoxCox applied training data

```
#The first difference will be applied to the train_bc data as before.
train_bc_1d<-diff(train_bc)
#After the BoxCox Transformation, it looks lot better.
autoplot(train_bc_1d,col="royalblue")
```



Interpretation:

This graph shows that, the time series data from 2014 to 2019 i.e., **the training became stationary after applying BoxCox transformation without huge variances**. So, look at the below code where we are applying the **ndiffs() function** again to ensure that no seasonality is has appeared and you can see the output along with code which displayed **'0'** ,which means the data is **stationary**.

```
#Using the ndiffs() function,
#We'll double check even though the differenced graphic no longer appears to have any seasonal characteristics.
print(ndiffs(train_bc_1d))

[1] 0
```

- Ensuring that the data is stationary or not

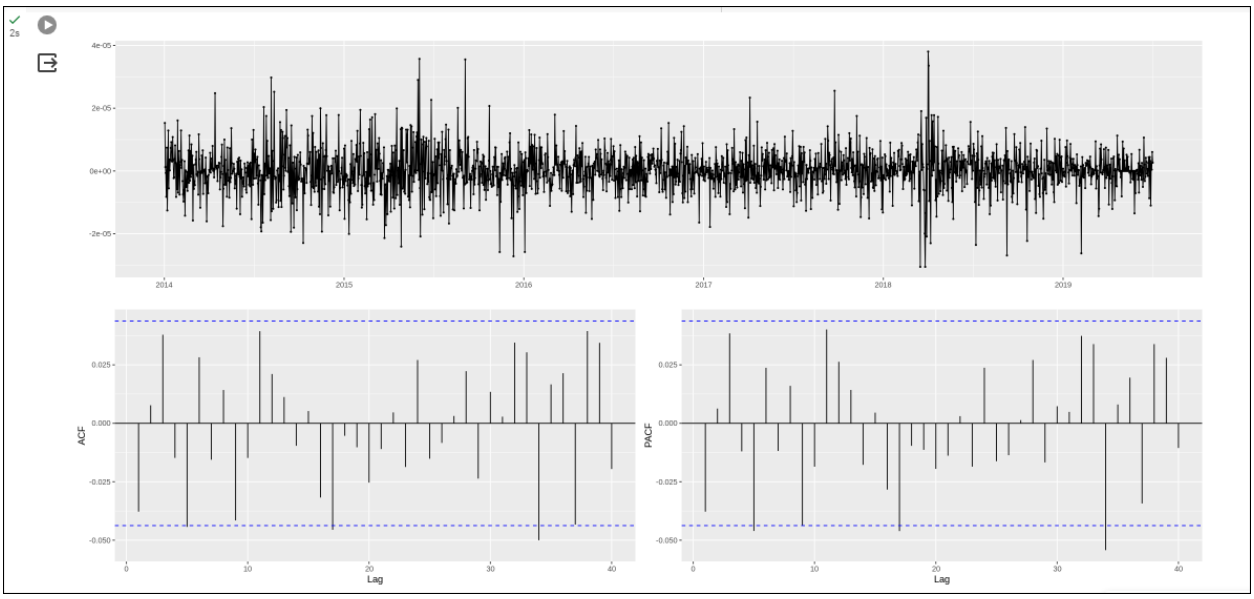
```
#Augmented Dickey-Fuller Test will be used on differenced data to make sure
library(tseries)
#Now that the p-value is sufficiently minimal (<0.05), we can confirm that the data is stationary.
adf.test(train_bc_1d)
```

```
Warning message in adf.test(train_bc_1d):
“p-value smaller than printed p-value”

        Augmented Dickey-Fuller Test

data:  train_bc_1d
Dickey-Fuller = -12.036, Lag order = 12, p-value = 0.01
alternative hypothesis: stationary
```

Interpretation:

Here, we have used **Augmented Dickery-Fuller Test** to ensure the stationarity. We got p-value as 0.01 which is less than actual p-value 0.05 i.e., **0.01<0.05**. So,we can reject **null hypothesis: the series has unit root**, that is we accept **alternative hypothesis: stationary**. Eventually,we can conclude that the time series data is stationary.

```
#At this point, ACF and PACF plots should be used to verify AR and MA degrees.
options(repr.plot.width=18,repr.plot.height=9)
# for about 2 out of 40 lags may be more than the margin, and it appears that all of the AR and MA degrees may be 0.
#In this case, ARIMA(0,1,0) may be an effective model.
ggtsdisplay(train_bc_1d,lag.max=40)
```



Interpretation

From the above graph, the **ACF** and **PACF** plots show that there is little evidence of significant correlation or partial correlation between observations at different time lags. Most bars in both plots fall inside the confidence interval, indicating **no strong autoregressive (AR)** or **moving average (MA)** effect. It suggests that both **AR** and **MA** degrees may be **0**, making **ARIMA(0,1,0) a potentially effective** model for the data. This model captures only the differencing component without requiring autoregressive or moving average terms.

- Here we have written code to see what model **auto.arima()** function suggests, So it has suggested the **ARIMA(1,1,0)**.

```
#let's observe the auto.arima() function,
#the ARIMA(1,1,0) model is suggested.
auto.arima(train_bc)
```

```
Series: train_bc
ARIMA(1,1,0)

Coefficients:
          ar1
      -0.0373
s.e.   0.0223

sigma^2 = 5.427e-10:  log likelihood = 21062.5
AIC=-42121.01   AICc=-42121   BIC=-42109.8
```

- Creating two models ARIMA(0,1,0) and ARIMA(1,1,0)

```
#Thus, we'll develop two models ARIMA(0,1,0) and ARIMA(1,1,0).
arima_010<-Arima(train_bc,order=c(0,1,0))
arima_110<-Arima(train_bc,order=c(1,1,0))
```

- Summary of arima_010 model

```
[34] summary(arima_010)
```

```
Series: train_bc
ARIMA(0,1,0)

sigma^2 = 5.425e-10:  log likelihood = 21061.11
AIC=-42120.21   AICc=-42120.21   BIC=-42114.61

Training set error measures:
                       ME         RMSE          MAE          MPE        MAPE
Training set 6.272505e-07 2.328527e-05 5.227506e-06 6.276707e-05 0.000523104
                     MASE       ACF1
Training set 0.06035531 0.01079299
```

- Summary of arima_010 model

```
summary(arima_110)

Series: train_bc
ARIMA(1,1,0)

Coefficients:
          ar1
      -0.0373
s.e.   0.0223

sigma^2 = 5.427e-10:  log likelihood = 21062.5
AIC=-42121.01   AICc=-42121   BIC=-42109.8

Training set error measures:
                       ME         RMSE          MAE          MPE         MAPE
Training set 6.320545e-07 2.328392e-05 5.223003e-06 6.324768e-05 0.0005226532
                    MASE       ACF1
Training set 0.06030331 0.01392941
```

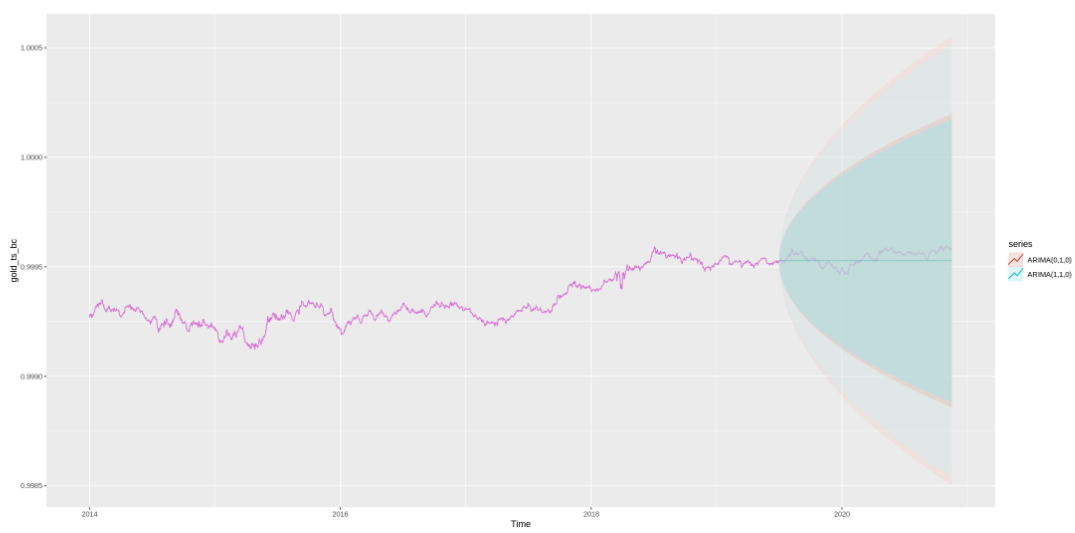
Forecasting

- We are forecasting the remaining data which we have not used for training

```
[36] #we'll forecast the next 503 days.
     NROW(test_bc)

     503

[37] arima_010_pred<-forecast(arima_010,h=503)

[38] arima_110_pred<-forecast(arima_110,h=503)

[40] #To plot them all at once, we'll apply the same BoxCox Transformation to the original data.
     gold_ts_bc<-BoxCox(gold_ts,lambda=train_lambda)
```

```
[41] #The ARIMA(1,1,0) model appears to be somewhat better to the ARIMA(0,1,0) model.
     autoplot(gold_ts_bc,col="orchid")+autolayer(arima_010_pred,series="ARIMA(0,1,0)",alpha=.3)+autolayer(arima_110_pred,series="ARIMA(1,1,0)",alpha=.3)
```

[41]



Interpretation:

From the above visualization, we can figure out that the arima(0,1,0) model performed better compared to arima(1,1,0). But, to understand more clearly let's compare the metrics of two models.

- Comparing the Accuracy of two models arima(0,1,0) and arima(1,1,0), to verify which model performs better.

```
#We'll Compare two models with metrics such as ME,RMSE,MAE,etc.
accuracy(arima_010_pred,test_bc)
accuracy(arima_110_pred,test_bc)
```

A matrix: 2 × 8 of type dbl

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1	Theil's U
Training set	6.272505e-07	2.328527e-05	5.227506e-06	6.276707e-05	0.000523104	0.06035531	0.01079299	NA
Test set	1.551365e-05	3.473946e-05	2.982408e-05	1.551977e-03	0.002983751	0.34434041	0.98564418	7.072923

A matrix: 2 × 8 of type dbl

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1	Theil's U
Training set	6.320545e-07	2.328392e-05	5.223003e-06	6.324768e-05	0.0005226532	0.06030331	0.01392941	NA
Test set	1.560905e-05	3.478217e-05	2.986219e-05	1.561521e-03	0.0029875635	0.34478043	0.98564429	7.081618

Interpretation:

Based on the provided accuracy values, the ARIMA(0,1,0) model performs better than the ARIMA(1,1,0) model. The ARIMA(0,1,0) model exhibits lower values for metrics such as RMSE, MAE, MPE, MAPE, and MASE on both the training and test sets. This indicates that the ARIMA(0,1,0) model has better accuracy and is closer to the actual values compared to the ARIMA(1,1,0) model. Therefore, in terms of forecast accuracy, the ARIMA(0,1,0) model is the preferable choice.

6. CONCLUSION

In today's world gold has taken the major part for investment and day by day the value of the gold increasing a lot in market.

- **Trend of Gold Price:** We found that the price of the gold has been rising up over the past decade which means gold is valuable and plays significant role in this world.
- **Best Model:** For forecasting the gold price of 503 days from 184th day of 2019 to Jan 2024, we have performed two models which are ARIMA(0,1,0) and ARIMA(1,1,0). In this two models we found that the ARIMA(0,1,0) performed better than ARIMA(1,1,0). Which means ARIMA(0,1,0) made accurate predictions.
- **Better Accuracy:** ARIMA(0,1,0) model forecasted well which the observations are closer to the actual gold price. Moreover, it has lower errors in prediction compared to ARIMA(1,1,0) model.
- **Investment Insights:** This study demonstrates how statistical tools such as ARIMA can help investors and others better comprehending and predicting gold prices. Knowing how gold prices have fluctuated in the past allows people to make greater decisions when buying, selling, or investing in gold.

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